

Module 3 — Thursday Stretch: Time-Series Analytics with Window Functions

Honors Track. This stretch assignment is not required for program completion but counts toward Honors distinction. Only work on this if you have completed all core assignments for this module and are On Track or Advanced. If you are behind on core work, focus there first.

The Scenario

You are a Business Intelligence (BI) analyst at a mid-size e-commerce company. The Vice President (VP) of Revenue has asked for a quarterly executive report covering revenue trends, customer retention, and product category performance over the past 12 months. Your analysis will inform budget allocation and acquisition strategy for the next quarter.

Your tool: advanced SQL window functions — the same techniques BI teams use to produce the period-over-period comparisons, cohort analyses, and trend lines that drive real business decisions.

What You'll Build

- **Four SQL analysis files** using window functions you haven't practiced in core assignments: LAG/LEAD for period-over-period growth, frame specifications for moving averages, ROW_NUMBER for cohort definition, and combined multi-window queries
- **An executive report** (`report.md`) summarizing your findings for a non-technical audience — claims supported by data, with actionable recommendations

BI analysts use the skills you'll use today daily to answer questions like “why did revenue change last quarter?” and “which customer segments should we invest in?” You are learning the ability to write these queries and then translate the results into a clear recommendation for someone who doesn't read SQL. Build each query yourself, understand what every clause does, and be ready to explain your reasoning.

Do not rely on AI to interpret the data or write your analysis. Pattern recognition, business judgment, and analytical narrative are the skills this assignment is designed to build, and they are what you will need in an interview. So be sure to do the thinking yourself.

What You'll Learn

- Using LAG and LEAD to compute period-over-period growth rates (month-over-month, quarter-over-quarter)
- Writing frame specifications (ROWS BETWEEN) for moving averages that smooth out noise
- Defining customer cohorts with ROW_NUMBER and analyzing retention across cohorts
- Composing multiple window functions in a single query to answer complex business questions
- Translating SQL output into business insight — the skill that separates analysts from query writers

Setup

Accept the [GitHub Classroom assignment](#) and clone your repo.

The `data/` directory contains CSV files and a `schema.sql` for four tables:

Table	Description
<code>customers</code>	5,000 customers — <code>customer_id</code> , <code>signup_date</code> , <code>segment</code>
<code>products</code>	300 products across 6 categories — <code>product_id</code> , <code>name</code> , <code>category</code> , <code>unit_price</code>
<code>orders</code>	~30,000 orders over 12 months — <code>order_id</code> , <code>customer_id</code> , <code>order_date</code> , <code>status</code>
<code>order_items</code>	~68,000 line items — <code>order_item_id</code> , <code>order_id</code> , <code>product_id</code> , <code>quantity</code> , <code>unit_price</code>

Create a PostgreSQL database and load the data. If you need a refresher, refer to the **Setup** section of the Lab 3 Guide.

One thing to note: `order_items.unit_price` is the price at time of purchase. It may differ from the product's list price in the `products` table.

Tasks

1. Cohort Analysis — `queries/cohort_analysis.sql`

Define customer cohorts by first-purchase month using ROW_NUMBER (not by signup date — a customer who signed up in April but first purchased in June belongs to the June cohort). Then analyze how cohorts behave over time:

- How many customers are in each cohort?
- What percentage of each cohort makes a repeat purchase within 30, 60, and 90 days?
- Which cohorts have the strongest retention? The weakest?

Your queries should use window functions to identify each customer's first purchase and to compute retention metrics across cohorts.

2. Growth Analysis — `queries/growth_analysis.sql`

Use LAG and LEAD to compute period-over-period comparisons:

- Month-over-month revenue growth rate
- Month-over-month order volume growth rate
- Quarter-over-quarter revenue growth rate

Look at the results critically. If revenue grew 15% one quarter, *why*? Is it more customers, higher order values, category mix, or something else? Your SQL should help answer this — not just compute the top-line number.

3. Trend Analysis — `queries/trend_analysis.sql`

Calculate moving averages using window frame specifications:

- 7-day moving average of daily revenue
- 30-day moving average of daily revenue
- 7-day moving average of daily order count

Use `ROWS BETWEEN` to define your window frames explicitly. Compare the raw daily numbers to the smoothed trend lines. Where do the moving averages reveal patterns that the daily data obscures?

4. Combined Analysis — `queries/combined_analysis.sql`

Write at least two queries that combine multiple window functions in a single query. This is the hardest part — it requires you to think about how different analytical lenses interact. Examples of what a combined query might look like:

- Cohort retention rates with period-over-period change (ROW_NUMBER + LAG)
- Category revenue share with moving average trend (SUM window + frame spec)
- Monthly revenue by segment with growth rate and running total (LAG + SUM window)

These queries should surface something that no single-window query could show on its own.

5. Executive Report — `report.md`

Write a business-facing executive report. This is your primary deliverable — the queries are the engine; the report is what leadership reads. Structure it with these sections:

- **Revenue Trends:** What happened over the 12 months? Where is growth coming from? Are there anomalies?
- **Customer Retention:** Which cohorts retain best? What does this suggest about acquisition channels or timing?
- **Category Performance:** Which categories are growing or declining? What does the mix shift mean for the business?
- **Recommendations:** Based on your analysis, what should the business do next quarter?

Support every claim with specific numbers from your queries. A good executive report makes the data tell a story — it doesn't just list results.

What the Autograder Does and Does Not Check

The CI workflow runs when you open your PR. It validates:

- Required SQL files exist in `queries/`
- Each SQL file executes without errors against the dataset
- Each SQL file contains at least one window function (`OVER` clause)
- `report.md` exists and includes the required sections

Green checks confirm your files are structurally sound. They do not evaluate the quality of your analysis. All points for this assignment are based on analytical depth, query sophistication, and report quality.

Portfolio Artifact

This stretch produces a strong portfolio piece. A polished executive report backed by sophisticated SQL demonstrates exactly the analytical communication that hiring managers look for in BI and analytics roles. Your queries show technical depth; your report shows business judgment.

Submit

1. Create a branch (e.g., `stretch-time-series`)
2. Add your query files and `report.md`
3. Open a PR to `main`
4. Paste your PR URL into TalentLMS → Module 3 → Thursday Stretch to submit this assignment

